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Residual Load and Changing Penetration Levels

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09/07/2026

Introduction and Research Objective

Research Objective

How do changing renewable penetration levels affect the residual load profile?

Residual Load

Residual load is defined as:

$$R_t = D_t - \alpha_s C_s \rho_{s,t} - \alpha_w C_w \rho_{w,t}$$

R_t = residual load, D_t = demand, C_s, C_w = 100%-layout solar/wind capacity, $\rho_{s,t}, \rho_{w,t}$ = relative production, α_s, α_w = penetration scaling

It represents the demand left over after solar and wind generation have been accounted for.

Choice of Penetration Levels

2011: $\alpha_{\text{Solar}} = 1\%$, $\alpha_{\text{Wind}} = 5\%$

Levels at beginning of RE-Europe dataset;

2025: $\alpha_{\text{Solar}} = 13\%$, $\alpha_{\text{Wind}} = 17\%$

2025 penetration levels per Ember 'European Electricity Report'. [1]

2030: $\alpha_{\text{Solar}} = 22\%$, $\alpha_{\text{Wind}} = 28\%$

Estimate of penetration levels needed to achieve the EU Renewable Energy Directive (RED) III Target of 42.5%. [2]

Wind and solar penetration in EU electricity

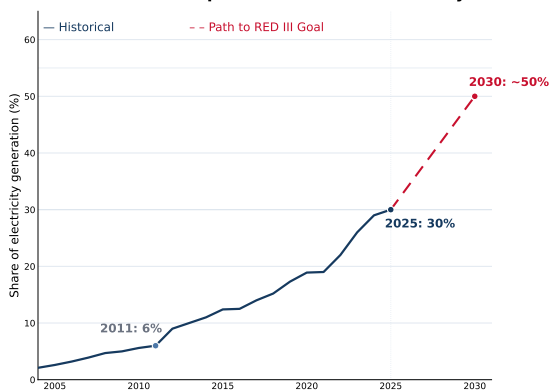
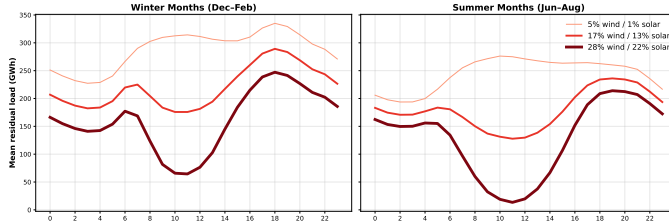


Figure: Historical levels taken from Ember European electricity report. Future estimates based on levels needed to reach RED III Target of 42.5% for all energy usage

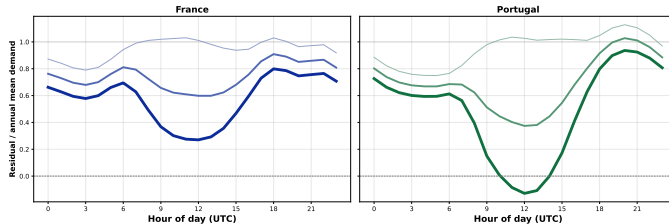
Typical Residual Load Behaviour

Characterising the diurnal & seasonal structure of residual load across countries

(A): Average diurnal residual load by season



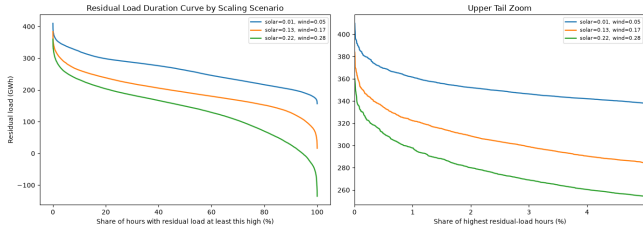
(B): Diurnal residual load by country



- Diurnal shape is driven by a midday solar dip and a sharp evening demand ramp.
- Winter accentuates the evening peak; summer deepens the midday trough.
- Higher penetration affects countries differently depending on their generation mix – Portugal and France shown here as one contrast.

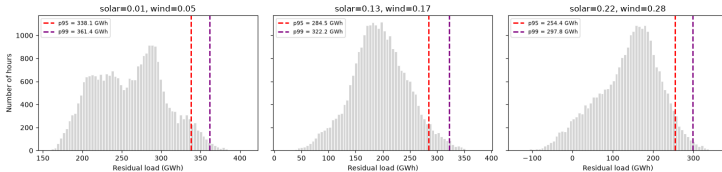
Residual Load Tail Risk

Impact of solar and wind scaling on upper-tail residual load



(A) : Duration curve

Residual Load Distribution and Upper-Tail Thresholds by Scaling Scenario



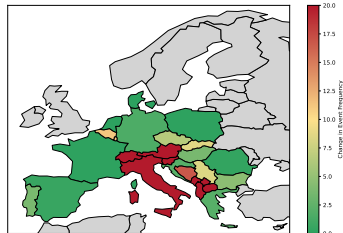
(B) : Residual-load distribution

- Higher solar/wind scaling increases renewable contribution and lowers residual load.
- The duration curve shifts downward across scenarios, especially in the upper tail.
- p95 and p99 thresholds fall as scaling increases.
- Tail risk is reduced but not eliminated, so backup capacity or flexibility is still needed.

Extreme Residual Load Distribution

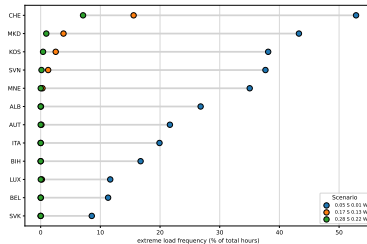
Impact of solar and wind scaling on Extreme Event distribution

Extreme events defined to be events in the top 5% by residual load over demand across all scenarios.

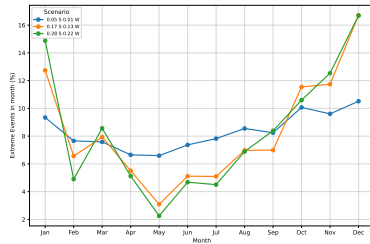


Change in extreme event frequency per country.

$$\Delta \text{pct}_{\text{extreme}} = \text{pct}_{\text{extreme}}^{(2030)} - \text{pct}_{\text{extreme}}^{(2011)}$$



Extreme event frequency across different penetration scenarios. Countries shown had the most extreme events in 2011.



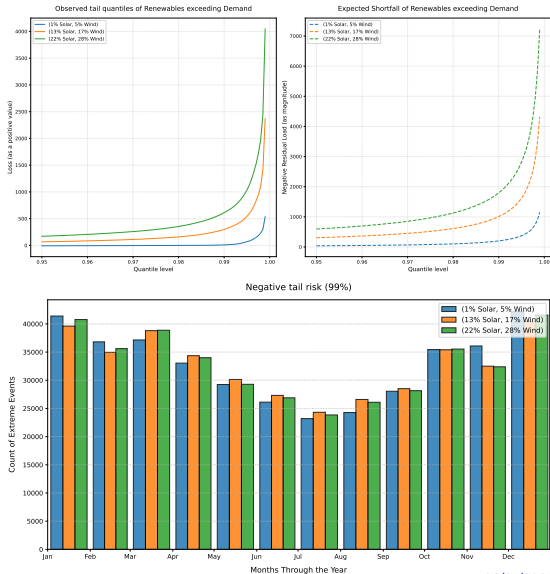
Monthly distribution of extreme events across different penetration scenarios.

Extreme negative residual load

Measures of extreme risk

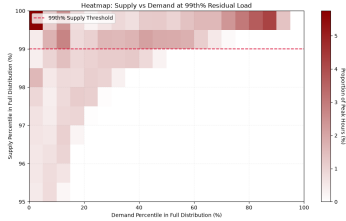
Quantile evolution from 0.95 to 0.999, highlights that beyond 0.995% the magnitude of extreme values increases significantly.

The expected shortfall graphic suggest that breach beyond the 99.5% threshold will likely cause massive oversupply of energy.

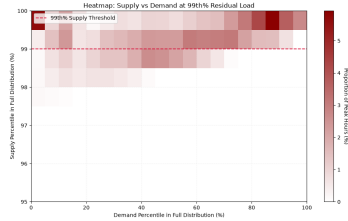


Extreme Surplus Energy

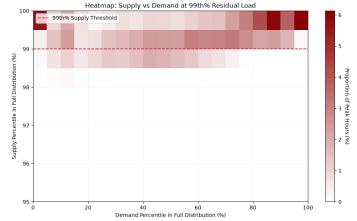
Supply v Demand Contribution



For this penetration level (0.01,0.05), the 99th percentile of supply is $\approx 125\text{MWh}$.



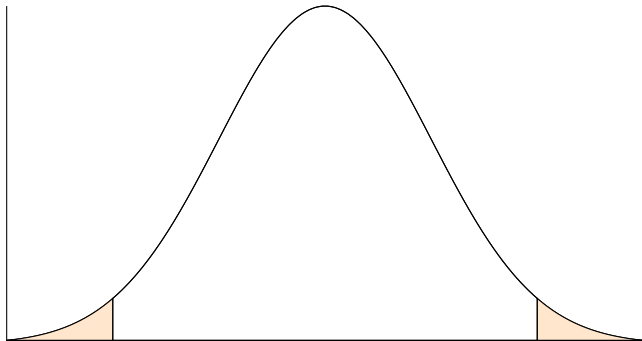
For this penetration level (0.13,0.17), the 99th percentile of supply is $\approx 500\text{MWh}$.



For this penetration level (0.22,0.28), the 99th percentile of supply is $\approx 1000\text{MWh}$.

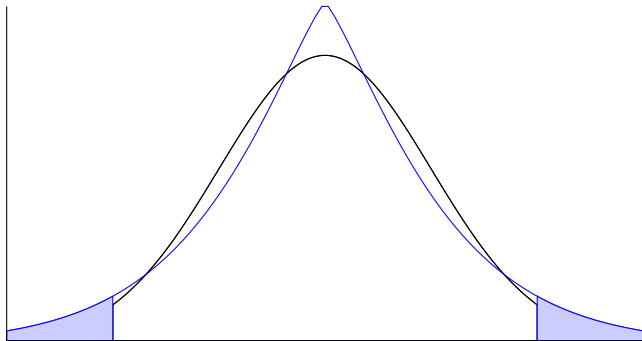
What are useful predictors of stress events?

ramp rate distribution tails



What are useful predictors of stress events?

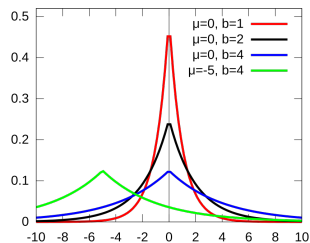
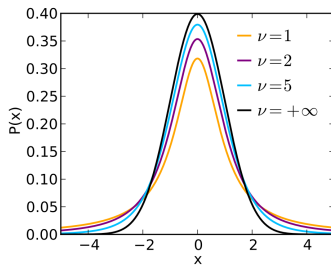
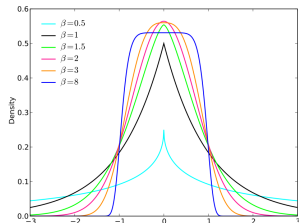
capturing tail weights is crucial!



Distribution Fitting

Distribution	$\alpha_s = 0.01, \alpha_w = 0.05$	$\alpha_s = 0.13, \alpha_w = 0.17$	$\alpha_s = 0.22, \alpha_w = 0.28$
Generalised Normal	21 (84%)	20 (80%)	22 (88%)
Student's t	4 (16%)	5 (20%)	2 (8%)
Laplace	–	–	1 (4%)
Total	25	25	25

Table: best-fit distribution tally across countries; initial goodness of fit ranked via AIC and verified with the K-S test



Distribution Parameters

Distribution	Parameter	$\alpha_s = 0.01, \alpha_w = 0.05$	$\alpha_s = 0.13, \alpha_w = 0.17$	$\alpha_s = 0.22, \alpha_w = 0.28$
Generalised Normal	shape β	0.6283 (0.02169)	0.6345 (0.0221)	0.5577 (0.02694)
	loc μ	-0.2336 (0.2162)	-0.6316 (0.3689)	-0.8182 (0.4323)
	scale α	2.618 (4.005)	5.177 (12.09)	4.978 (14.94)
Student's t	df ν	1.293 (0.02338)	2.132 (0.8499)	1.462 (0.4638)
	loc μ	-0.3883 (0.134)	-0.5389 (0.1208)	-0.8296 (0.0104)
	scale τ	3.361 (2.098)	9.438 (5.519)	11.65 (3.462)
Laplace	loc μ	-	-	-0.166 (-)
	scale b	-	-	16.54 (-)

Table: Fitted parameters across runs (mean (variance))

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Thank you

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Ember.

European electricity review 2026.

Technical report, Ember, 2026.

EU electricity generation and demand data for 2025.



European Parliament and Council of the European Union.

Directive (eu) 2023/2413 of the european parliament and of the council of 18 october 2023 as regards the promotion of energy from renewable sources.

Official Journal of the European Union, 2023.

Amending Directive (EU) 2018/2001, Regulation (EU) 2018/1999 and Directive 98/70/EC.